Dairy, Agriculture & Finance Integrated Platform
Project Proposal (Phase-wise, v3 – with Financial Analysis & Stage Dependencies)

Summary
Design and develop a multi-phase, integrated Farm Management Platform to manage:
- Financial management (budgets, transactions, cashflow, sales projections, company valuation, financial analysis & secureness)
- Dairy farming operations (from calf birth through heifer, pregnancy, milking, dry period, to death)
- Agriculture operations (land, plot allocation, crop planning, yields)
- Warehouse & processing (stock and product processing flows)
- CRM / Sales & Marketing (orders, dispatch, payment & feedback)
- Planning & Operations (manual demand and capacity planning)

The platform will use a React frontend, Node.js backend (REST API), and SQL database, with a shared Master Data Management (MDM) layer to keep all modules on a common, standardised data structure.

Project Outline (Phase-wise)

Phase 1 – Financial Management
Objective:
Build the core finance engine: budgeting, financial transactions, cashflow monitoring, company valuation, financial analysis, and basic sales projections. This becomes the financial backbone for all later modules.

Scope:
- MDM_Finance & Shared Masters (part of global MDM)
- MDM_Product (product master: raw milk, pasteurised milk variants, ghee, cheese, fodder, crops, etc.)
- MDM_Account (bank, cash, wallets)
- MDM_TxnCategory hierarchy as per required field structure
- MDM_Customer (for sales revenue and projections)
- MDM_Vendor (for purchases)
- MDM_Asset (asset master, with asset type and category)
- MDM_CostCenter (e.g., Dairy, Agri, Admin, Warehouse, CRM, Processing)
- MDM_OperationUnit (e.g., Cow, Acre, Litre, Kg – used for operational cost per unit)

- Financial Field Structure (for all budgets and transactions)
- TxGroupName (Picklist: Investment, Expense, Income)
- TxSubgroupName (Picklist with dependency on TxGroupName; e.g., Investment: Capital Investment; Expense: CAPEX, Operational Cost; Income: Sales)
- BudgetGroupName (Picklist, free list; configurable groups for reporting)
- CategoryName (Picklist; e.g., Feed, Labour, Utilities, Veterinary, Fuel, etc.)
- SubCategoryName (Picklist; detailed break-up under each Category)
- ItemName (Single line text; specific item or purpose)
- UOM (Unit of Measure; e.g., kg, litre, piece)
- AssetType (Link/Picklist mapped to MDM_Asset; e.g., Land, Building, Livestock, Equipment)
- AssetCategory (Further classification under AssetType; e.g., Tractor, Chiller, Shed, Cow, etc.)
- CostCenter (Picklist linked to MDM_CostCenter; e.g., Dairy, Agri, Admin)
- OperationUnitType (Picklist linked to MDM_OperationUnit; identifies main unit for analysis, e.g., Cow, Acre, Litre)
- OperationUnitRef (Optional reference ID to link a transaction to a specific herd group, field, batch, etc.)

- Budgeting
- Define yearly / monthly budgets per TxGroupName, CategoryName, SubCategoryName and optionally per Product, Asset, CostCenter and OperationUnitType.
- Support expense, income, and investment budgets using the above field hierarchy.
- Prepare structure for later “budget vs actual” reporting and planning.

- Financial Transactions
- Record income, expenses, transfers, and investments using the full field structure: TxGroupName, TxSubgroupName, BudgetGroupName, CategoryName, SubCategoryName, ItemName, UOM, AssetType, AssetCategory, CostCenter, OperationUnitType, OperationUnitRef, Account, Product (if

relevant), Customer/Vendor, and optional operational reference.
- Each transaction will be mapped to relevant assets from the master and cost centres where applicable.
- Provide filterable lists and basic reports.

- Cash Flow Monitoring
- Monthly/weekly cash in/out.
- Closing balance by account.
- Combined cash flow chart.

- Company Valuation (Monthly)
- Compute valuation by combining:
- Assets (Land, Buildings, Livestock, Equipment, Inventory) from Asset master and ledger.
- Fund balances (Cash, Bank accounts).
- Generate monthly valuation snapshots for tracking growth over time.

- Financial Analysis & Secureness
- Operational Cost (Business Level Summary)
- Calculate total monthly operational cost by CostCenter (Dairy, Agri, Admin).
- Dairy: derive cost per cow per month by splitting expenses (fodder, healthcare, labour, utilities) and dividing by total active cows from Dairy module.
- Agri: derive cost per acre per month by summing Agri-related expenses and dividing by total cultivated acres from Agri module.
- Operational Cost (Product Level)
- Compute cost per UOM (e.g., per litre, per kg, per piece) for key products such as milk, cheese, ghee, curd, etc.
- Raw milk cost includes only farming operational cost (Dairy + directly related Agri fodder cost).
- Processed products (ghee, cheese) include farming cost + processing cost (from Warehouse/Processing module).
- Cash Flow Secureness (Operational)
- Determine how many months the business can continue operations with current fund balance (Cash + Bank) based on current monthly operational cost.
- Monthly operational cost is calculated from actual transactions linked to CostCenters (Dairy, Agri, Admin) and OperationUnitType.
- Cash Flow Secureness (Operational + Investments)
- Extend secureness calculation to include both operational cost and planned/average investment (CAPEX) cost.
- Provide separate indicators: “Operational only” and “Operational + Investment” months of secureness.

Deliverables:
- Finance schema (SQL tables) and API endpoints.
- React pages for budgets, transactions, cashflow, valuation view, and financial analysis dashboards.
- Seed data for key masters (accounts, basic products, categories, assets, cost centres).
- Documentation for finance workflows, mapping rules, and financial analysis logic.

Phase 2 – Dairy Farming Operations / Tracking
Objective:
Track the entire dairy livestock lifecycle, from calf birth to death, including all stages (calf, heifer, pregnant, milking, dry, dead), with stage-specific events, dependencies, manual/auto triggers, milk yield, and fodder and health requirements linked to finance via MDM.

Scope:
- MDM_Dairy (part of global MDM)
- MDM_Breed (breed-specific traits, expected yield, typical lactation length).
- MDM_CowStatus / LifecycleStage:
- Calf (0–X months)
- Heifer (growth stage before first calving)
- Pregnant (first pregnancy and later pregnancies)
- Milking (lactating cow, with sub-stages)
- Dry (dry period before calving)
- Dead / Culled / Sold (terminal stages)

- MDM_DairyLocation (sheds, pens, paddocks)
- MDM_DairyEvent (event master with stage mapping, dependencies, and mandatory/optional flags)
- Examples: Calf registration, Weaning, Breeding/AI, Pregnancy check, Calving, Vaccination, Deworming, Hoof trimming, Dry-off, Post-partum check, Lactation start, Lactation end, etc.
- Dependency fields for events and stages:
- StageDependencyType (e.g., “Milking depends on Pregnant/Calving”, “Dry depends on Milking end”).
- EventDependencyType (e.g., “Post-delivery vaccination depends on Calving event completion”).
- DependencyMode (Manual / Auto trigger based on dependency completion).
- Use MDM_Product for milk-related products (raw, chilled, pasteurised, packet variants).

- Lifecycle & Stage Management
- Define the cow lifecycle as a sequence of stages from calf to death.
- Each stage has:
- A typical duration in days (configurable per breed).
- A set of events linked to that stage, with dependencies.
- Trigger rules (manual or automatic based on dates/conditions).
- Example milking sub-stages by lactation days:
- Early lactation: 0–90 days
- Mid lactation: 91–150 days
- Late lactation: 151–240 days (values configurable per breed).
- Stage dependencies:
- Milking stage is dependent on Pregnant/Calving events being completed.
- Dry period is dependent on end of Milking sub-stages or planned dry-off event.
- For each stage and sub-stage:
- Expected milk yield range (for monitoring performance).
- Fodder requirement ranges or ration templates.
- Health monitoring focus (e.g., post-partum health checks, metabolic disorders risk).

- Events Master, Dependencies & Trigger Types
- Each event in MDM_DairyEvent is mapped to:
- One lifecycle stage or multiple stages.
- One or more breeds (if breed-specific).
- Trigger type:
- Manual (user initiated)
- Automatic (system-generated based on date, stage, or dependency completion)
- Trigger classification:
- Mandatory (must be done and tracked)
- Optional (recommended but not enforced).
- Dependency examples:
- Post-delivery vaccination event is auto-triggered when a Calving event is completed (EventDependencyType).
- Dry-off event required a minimum number of days of milking or pregnancy status confirmation.
- Vaccinations during specific seasons may be Optional; health supplements post-pregnancy (post-partum) and dry-off procedures can be Mandatory.

- Stage Day Count & Event Logging
- For each livestock record, log:
- Current stage, stage start date, stage end date (when moved).
- Number of days spent in each stage.
- All events are timestamped and linked to:
- The cow
- Its current stage at the time of event
- Any dependency events/stages.
- Historical lifecycle view per animal (timeline of stages, events, and dependencies).

- Livestock Purchase & Master
- Register new animals (calves, heifers, cows) with source and purchase cost (linked to finance transaction).

- Track cow ID, tag number, breed, DOB/age, current lifecycle stage, status, and location.

- Milk Yield Tracking
- Log milk yield per cow or per group per milking session/day.
- Use lactation day number to classify yield into early/mid/late lactation.
- Link produced milk quantities to MDM_Product (raw milk) for downstream modules (warehouse/CRM).

- Fodder Requirement & Health Monitoring by Stage
- For each stage and breed:
- Define baseline fodder requirement table (kg DM/day or ration composition).
- Link to ration plans and feeding logs (later integration with Agri and Warehouse).
- Stage-specific health monitoring configuration:
- E.g., increased monitoring in early lactation and post-partum, targeted checks in dry period.

Deliverables:
- Extended Dairy schema and API endpoints (animals, stages, events, dependencies, lifecycle transitions, milk logs, stage-day counters).
- React pages:
- Herd list and filters by stage.
- Cow profile with lifecycle timeline, events and dependency indicators.
- Event management (master and per-cow events).
- Stage-specific monitoring screens.
- Integration with Finance for livestock purchase, major treatments, and valuation.

Phase 3 – Agri Farming Operations
Objective:
Build the agriculture layer which provides fodder and crops; link them to dairy operations and financial valuation.

Scope:
- MDM_Agri (part of global MDM)
- MDM_FieldType, MDM_Crop, MDM_Season.
- Use MDM_Product for agricultural products (e.g., “Maize Silage”, “Napier Green Fodder”, “Paddy Grain”).

- Land / Plot Master
- Register all land parcels with area, soil type, irrigation source, GPS/landmarks.
- Support plot-level subdivision where needed (Field → Subplots).

- Crop Planning
- Plan crop type, season, sowing date, and expected harvest date.
- Estimate expected yield per acre and total expected yield.
- Map which crops are destined for fodder vs market sale.

- Yield Recording
- Record actual harvest quantity and DM conversion for fodder.
- Link harvested output into warehouse/stock (Phase 4) and to MDM_Product.
- Provide data for financial valuation through Phase 1 finance module.

Deliverables:
- Agri schema and API endpoints (fields, crop plans, crop cycles, yields).
- React pages for land master, crop plans, yield summaries, and fodder availability.
- Integration to warehouse stock and finance cost/revenue where relevant.
- Acreage totals and cropping data made available to Finance module for operational cost per acre and capacity planning.

Phase 4 – Warehouse / Processing Operations
Objective:
Collect all outputs (from dairy & agri) into a unified stock/warehouse system and model processing flows from raw materials to finished products, which will feed product-level cost analysis in Finance.

Scope:
- MDM_Warehouse (part of global MDM)
- MDM_WarehouseLocation (cold room, fodder shed, finished goods store).
- MDM_ProcessType (Pasteurisation, Ghee Making, Cheese Making, etc.).
- All finished products defined in MDM_Product (Milk SKUs, Ghee SKUs, Cheese SKUs, etc).

- Raw Material Intake
- Raw milk intake from Dairy to Warehouse.
- Fodder/crop intake from Agri to Warehouse stock.
- Traceability via reference IDs (cow group/batch, crop cycle).

- Stock & Processing
- Maintain stock quantities of raw milk, semi-processed, and finished products.
- Model basic process flows:
- Raw milk → Pasteurised milk.
- Raw milk → Cream → Ghee / Butter / Cheese.
- Generate finished goods with cost basis from raw material + process cost (linked to Finance).

- Sales vs Processing Decision (at batch level)
- Option to mark incoming raw milk/crops as:
- Direct sales (goes to CRM/sales module).
- Processing (goes through process flows before sales).
- Processing cost data feeds into Finance product-level cost calculation.

Deliverables:
- Warehouse schema and API endpoints (stock, processes, movements).
- React pages for stock views, processing forms, and batch history.
- Integration with Dairy, Agri, Finance, and CRM modules.

Phase 5 – CRM (Sales / Marketing)
Objective:
Build sales & CRM layer linked to products and stock, with customer ordering, payment terms, payment type, and feedback, and ensure all sales data is available for financial and operational analysis.

Scope:
- Customer Management (MDM_Customer in MDM)
- Customer master for retail, B2B, distributors (with type/segment).

- Customer Order Capture
- Provide interface for customers (or internal user on behalf of customer) to place orders from product master (MDM_Product).
- Support order types by customer segment (retail, business, bulk, distributor).

- Order Processing & Sales Order Conversion
- Received customer orders can be:
- Manually approved and converted to Sales Order.
- Auto-approved based on configurable rules (e.g., credit status, standard products).
- Once approved, the system generates Sales Order records that feed into warehouse dispatch and finance invoicing.

- Payment Terms & Payment Type
- Each order to capture customer-specific payment terms, e.g.:
- Advance (100% before delivery)
- 50/50 (50% advance, 50% on delivery)
- Post-delivery net terms (e.g., 7 days, 15 days, 30 days)
- Payment type fields:
- Cash, Bank transfer, UPI, Cheque, or others (configurable).
- These fields flow into Finance module for correct transaction and credit tracking.

- Sales Tracking, Invoicing & Payment
- Track:
- Order status (New, Approved, Dispatched, Delivered, Invoiced, Paid, Partially Paid, Overdue).
- Dispatch details (date, vehicle, route, batch).
- Generate invoices (internal format) tied to sales orders and products.
- Record payments against invoices:
- Full, partial, or multiple payments.
- Option to record sales and payments digitally through app or mark as manual (offline) with summary captured in system.

- Customer Feedback
- Capture feedback for each order or periodically:
- Product quality, service, delivery punctuality.
- Ratings and comments.
- Feedback can be used in planning and quality review.

Deliverables:
- CRM schema and API endpoints (customers, orders, sales orders, invoices, payments, feedback).
- React pages for order intake, approvals, sales order management, invoice/payment tracking, and feedback entry.
- Integration with Warehouse (stock allocation and dispatch) and Finance (revenue & receivables).

Phase 6 – Planning / Operations
Objective:
Build a cross-module planning layer to map demand, production capacity, and operations – initially manual forecasting based on market demand, with a strong link to financial analysis (operational and investment secureness).

Scope:
- Demand Planning
- Manual input of forecast demand per product (quantity/time period).
- Optionally segment by channel/customer.

- Production & Resource Planning
- Map forecast demand against:
- Dairy production capacity (milk yield and herd structure, by lifecycle stage and expected calvings).
- Agri fodder and crop availability (per season, per acre).
- Warehouse/processing capacity and existing stock.

- Operations Planning Views
- Show capacity vs demand:
- Demand > capacity $\rightarrow$ need to increase production or adjust sales.
- Capacity > demand $\rightarrow$ risk of wastage or need for new markets.
- Highlight stage-based requirements for dairy (e.g., future heifers, expected calvings) and required fodder cropping.
- Integrate outputs from financial analysis:
- Show impact on operational cost per cow/acre and product.
- Show updated cash flow secureness based on planned production and investments.

- Initial Manual Workflow
- Forecasts entered manually.
- Simple comparison dashboards; later can be extended to semi-automatic planning using historical data.

Deliverables:
- Planning schema and API endpoints (forecasts, capacity snapshots).
- React pages for planning input and dashboards.
- Integrations to pull data from Finance, Dairy, Agri, Warehouse, and CRM.

Apps Functionality Brief (Full System)
- User login and role-based access (Admin, Finance, Dairy, Agri, Warehouse, Sales, Planning, Viewer).

- Finance:
- Budgets, transactions, cashflow, sales projections, company valuation, and detailed financial analysis (operational costs, product-level costs, cash flow secureness).
- Dairy:
- Herd master, full lifecycle stage management from calf to death, stage-based events and dependencies, auto/manual triggers, milk yield recording, and stage-specific fodder and health monitoring.
- Agri:
- Land/plot master, crop plans, harvest & yield tracking, fodder supply to dairy, linkage to cost-per-acre analysis.
- Warehouse:
- Stock intake from Dairy and Agri, processing flows, finished goods management, processing cost data for Finance.
- CRM:
- Customer orders from product master, sales order conversion, payment terms, payment type, invoicing, payment tracking, and feedback.
- Planning:
- Manual demand forecasts, capacity vs demand comparison using integrated data from all modules; reference to financial analysis for decision-making.
- Common MDM:
- Central master data for Products, Customers, Vendors, Locations, Accounts, Categories, Breeds, Crops, Seasons, Process Types, Statuses, Assets, CostCenters, OperationUnits, and Events.

Technical Details
- Frontend:
- React SPA for all modules with modular routing per phase (Finance, Dairy, Agri, Warehouse, CRM, Planning).
- Shared components for tables, forms, filters, dashboards, and analysis views.
- State/data handling via React Query or equivalent.

- Backend:
- Node.js (Express or NestJS) REST API.
- Modular services per phase (finance, dairy, agri, warehouse, CRM, planning).
- Business rules and validations at the API layer, including:
- Lifecycle stage transitions, event dependencies, and automatic event generation in Dairy.
- Enforcing field structure, cost centre mapping, and financial analysis metrics in Finance.
- Order approval rules, payment term enforcement, and receivable tracking in CRM.

- Database:
- SQL database (PostgreSQL or MySQL).
- Normalised schema with a central MDM schema used by all modules.
- Proper foreign keys for traceability: e.g., transactions linked to operations events, orders linked to invoices and payments, dairy and agri data linked to financial analysis.

- Security & Deployment:
- JWT-based authentication and role-based authorization.
- Environment separation for development and production.
- API documentation using OpenAPI/Swagger.
- Cloud deployment for frontend, backend, and database with appropriate backups.